

Airline Operational Performance & Flight Delay Analysis with Big Data Analytics Using Python

1. Project Overview

Airline Operational Performance & Flight Delay Analytics is a data analytics project focused on evaluating airline operations, flight activity, and delay patterns using over 1 million flight records. The dataset was cleaned and analyzed using Python (Pandas, NumPy, Matplotlib, Seaborn) and visualized through interactive Power BI dashboards.

The project examines key operational metrics such as flight volume, airport traffic, taxi-in/taxi-out time, ground time, weather delays, and late aircraft delays to identify performance trends, operational bottlenecks, and delay drivers. The insights help assess airport efficiency, flight operations, and overall airline performance through data-driven decision-making.

2. Project Objectives

- Analyze airline operational performance using over 1 million flight records.
- Identify the busiest airports, states, and peak travel periods based on flight activity.
- Evaluate airport efficiency through taxi-in, taxi-out, and ground-time metrics.
- Examine the impact of weather delays and late aircraft delays on flight operations.
- Detect operational bottlenecks and congestion patterns affecting airline performance.
- Develop interactive Power BI dashboards to support data-driven operational decision-making.

3. Data Sources

a. Raw Dataset :

https://github.com/smalini369/Main_Project/blob/main/flight_data_2024.zip

b. Timeline : Data collected during Jan 2024 to Feb 2024

c. Cleaned Dataset :

https://github.com/smalini369/Main_Project/blob/main/Airline%20Operational%20Performance%20%26%20Flight%20Delay%20Analysis%20with%20Big%20Data%20Analytics%20Using%20Python_Cleaned.zip

d. Domain : Environment

4. Problem Statement

Airline operations involve managing large volumes of flight data across multiple airports, states, and time periods. Delays caused by weather conditions, airport congestion, and late-arriving aircraft can significantly impact operational efficiency, increase costs, and affect passenger satisfaction. Identifying the root causes of these delays is essential for improving airline performance and resource utilization.

This project aims to analyze airline operational and flight delay data to uncover traffic patterns, operational bottlenecks, and key delay drivers. By leveraging data cleaning, analysis, and interactive Power BI dashboards, the project provides actionable insights to support better decision-making, improve operational efficiency, and enhance overall airline performance.

5. Attribute (Column /Features) Details:

S.No	Attribute Name	Data Type	Description
1	year	Integer (Numeric)	Year of flight operation
2	month	Integer (Numeric)	Month number of the flight operation
3	day_of_month	Integer (Numeric)	Day of the month when the flight operated
4	day_of_week	Integer (Numeric)	Numeric representation of the day of the week
5	fl_date	Date/Time	Date of flight operation
6	day_name	String (Text)	Name of the day (Monday, Tuesday, etc.)
7	month_name	String (Text)	Name of the month (January, February, etc.)
8	origin	String (Text)	Departure airport code of the flight
9	origin_city	String (Text)	City where the departure airport is located
10	origin_state	String (Text)	State where the departure airport is located
11	dep_time	Time/Text	Actual departure time of the flight

Sl.No	Attribute Name	Data Type	Description
12	taxi_out	Integer (Numeric)	Time spent moving from gate to runway before takeoff (minutes)
13	taxi_in	Integer (Numeric)	Time spent moving from runway to gate after landing (minutes)
14	cancelled	Integer (Numeric)	Flight cancellation status (0 = Completed, 1 = Cancelled)
15	distance	Integer (Numeric)	Total distance traveled by the flight (miles)
16	weather_delay	Integer (Numeric)	Delay caused by weather conditions (minutes)
17	late_aircraft_delay	Integer (Numeric)	Delay caused by late arrival of the aircraft (minutes)
18	flight_status	String (Text)	Operational status of the flight (Completed / Cancelled)
19	weather_delay_log	Decimal (Numeric)	Log-transformed weather delay used to reduce skewness
20	ground_time	Integer (Numeric)	Total ground operation time (Taxi-Out + Taxi-In)
21	congestion_level	String (Text)	Categorized airport congestion level (Low, Medium, High)
22	time_slot	String (Text)	Categorized departure period (Morning, Afternoon, Evening, Night)

6. Tools & Technologies

1. **Python:** Data cleaning, data transformation, exploratory data analysis, and creating Pivot Tables.
2. **Power BI:** Data modelling, DAX calculations, data visualization, and interactive dashboard creation.
3. **Pandas & NumPy:** Data manipulation and numerical analysis.
4. **Matplotlib & Seaborn:** Data visualization and exploratory analysis.

Tasks Performed in Python

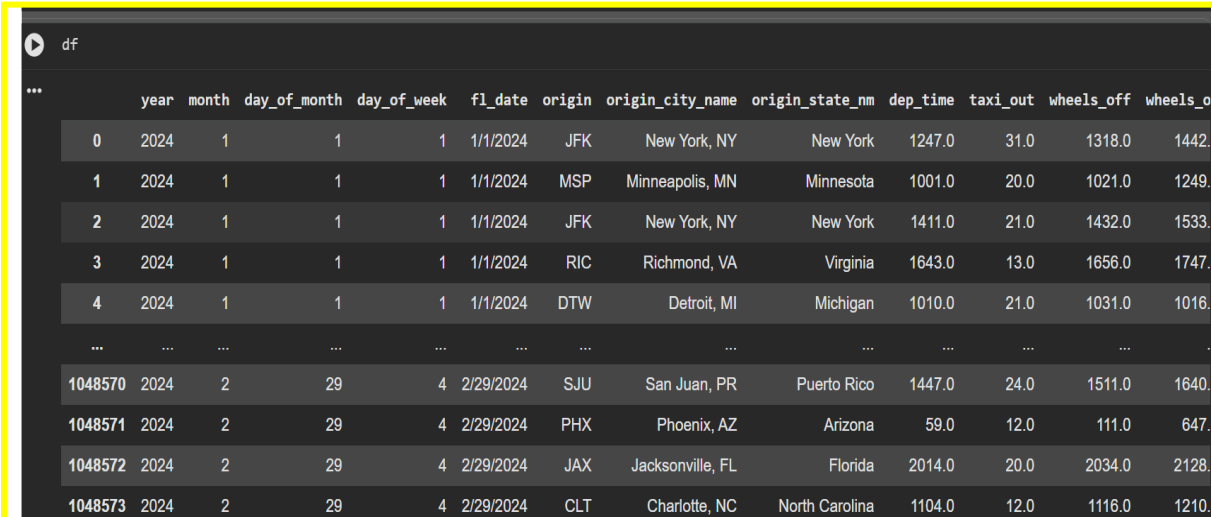
- Performed data cleaning and preprocessing using Pandas and NumPy.
- Handled missing values and validated data quality.
- Converted columns to appropriate data types.
- Treated outliers and reduced skewness using log transformation.
- Engineered new features such as day_name, month_name, flight_status, ground_time, congestion_level, time_slot, and weather_delay_log.
- Conducted Exploratory Data Analysis (EDA) using Matplotlib and Seaborn.
- Generated descriptive statistics and identified operational trends.

Tasks Performed in Power BI

- Imported and transformed the cleaned dataset using Power Query.
- Created data models and relationships for analysis.
- Developed DAX measures and KPI metrics.
- Built interactive dashboards for Airline Operations Performance & Flight Activity and Delay Diagnostics & Operational Performance.
- Created visualizations including KPI cards, bar charts, line charts, pie charts, gauge charts, and maps.
- Implemented slicers and filters for dynamic analysis.
- Derived insights related to flight activity, airport congestion, operational efficiency, and delay patterns.

7. Data Pre-Processing (Python / Power Query)

Initial EDA



	year	month	day_of_month	day_of_week	fl_date	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	wheels_off	wheels_o
0	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1247.0	31.0	1318.0	1442.
1	2024	1	1	1	1/1/2024	MSP	Minneapolis, MN	Minnesota	1001.0	20.0	1021.0	1249.
2	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1411.0	21.0	1432.0	1533.
3	2024	1	1	1	1/1/2024	RIC	Richmond, VA	Virginia	1643.0	13.0	1656.0	1747.
4	2024	1	1	1	1/1/2024	DTW	Detroit, MI	Michigan	1010.0	21.0	1031.0	1016.
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2024	2	29	4	2/29/2024	SJU	San Juan, PR	Puerto Rico	1447.0	24.0	1511.0	1640.
1048571	2024	2	29	4	2/29/2024	PHX	Phoenix, AZ	Arizona	59.0	12.0	111.0	647.
1048572	2024	2	29	4	2/29/2024	JAX	Jacksonville, FL	Florida	2014.0	20.0	2034.0	2128.
1048573	2024	2	29	4	2/29/2024	CLT	Charlotte, NC	North Carolina	1104.0	12.0	1116.0	1210.

df.head()

	year	month	day_of_month	day_of_week	fl_date	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	wheels_off	wheels_on	taxi_in
0	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1247.0	31.0	1318.0	1442.0	1442.0
1	2024	1	1	1	1/1/2024	MSP	Minneapolis, MN	Minnesota	1001.0	20.0	1021.0	1249.0	1249.0
2	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1411.0	21.0	1432.0	1533.0	1533.0
3	2024	1	1	1	1/1/2024	RIC	Richmond, VA	Virginia	1643.0	13.0	1656.0	1747.0	1747.0
4	2024	1	1	1	1/1/2024	DTW	Detroit, MI	Michigan	1010.0	21.0	1031.0	1016.0	1016.0

df.shape

(1048575, 18)

df.size

18874350

df.isnull().sum()

year	0
month	0
day_of_month	0
day_of_week	0
fl_date	0
origin	0
origin_city_name	0
origin_state_nm	0
dep_time	22553

taxi_out	23125
wheels_off	23125
wheels_on	23677
taxi_in	23677
cancelled	0
air_time	25751
distance	0
weather_delay	0
late_aircraft_delay	0
dtype:	int64
df.duplicated().sum()	
np.int64(7401)	

df.describe()

	year	month	day_of_month	day_of_week	dep_time	taxi_out	wheels_off	wheels_on	taxi_in	cancelled
count	1048575.0	1.048575e+06	1.048575e+06	1.048575e+06	1.026022e+06	1.025450e+06	1.025450e+06	1.024898e+06	1.024898e+06	1.048575e+06
mean	2024.0	1.478081e+00	1.530512e+01	3.893483e+00	1.325074e+03	1.825012e+01	1.349996e+03	1.476156e+03	8.082517e+00	2.222635e-02
std	0.0	4.995196e-01	8.585503e+00	2.010038e+00	4.972990e+02	1.044025e+01	4.980426e+02	5.198682e+02	6.512591e+00	1.474190e-01
min	2024.0	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	0.000000e+00
25%	2024.0	1.000000e+00	8.000000e+00	2.000000e+00	9.110000e+02	1.200000e+01	9.290000e+02	1.058000e+03	4.000000e+00	0.000000e+00
50%	2024.0	1.000000e+00	1.500000e+01	4.000000e+00	1.323000e+03	1.500000e+01	1.337000e+03	1.510000e+03	6.000000e+00	0.000000e+00
75%	2024.0	2.000000e+00	2.300000e+01	6.000000e+00	1.736000e+03	2.100000e+01	1.750000e+03	1.914000e+03	9.000000e+00	0.000000e+00
max	2024.0	2.000000e+00	3.100000e+01	7.000000e+00	2.400000e+03	2.130000e+02	2.400000e+03	2.400000e+03	4.440000e+02	1.000000e+00

Data Cleaning & Transformation

- Checked and handled missing values in the dataset.
- Verified data types and corrected inconsistent formats.
- Removed duplicate records to maintain data accuracy.
- Standardized categorical values and text formats.
- Converted date and time columns into suitable formats for analysis.
- Performed data transformation and created required fields for dashboard analysis.
- Prepared the cleaned dataset for Power BI visualization.

Handling Null Values:

```
df.drop_duplicates(inplace=True)
df
```

...	year	month	day_of_month	day_of_week	fl_date	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	wheels_off	wheels_o
0	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1247.0	31.0	1318.0	1442.
1	2024	1	1	1	1/1/2024	MSP	Minneapolis, MN	Minnesota	1001.0	20.0	1021.0	1249.
2	2024	1	1	1	1/1/2024	JFK	New York, NY	New York	1411.0	21.0	1432.0	1533.
3	2024	1	1	1	1/1/2024	RIC	Richmond, VA	Virginia	1643.0	13.0	1656.0	1747.
4	2024	1	1	1	1/1/2024	DTW	Detroit, MI	Michigan	1010.0	21.0	1031.0	1016.
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2024	2	29	4	2/29/2024	SJU	San Juan, PR	Puerto Rico	1447.0	24.0	1511.0	1640.
1048571	2024	2	29	4	2/29/2024	PHX	Phoenix, AZ	Arizona	59.0	12.0	111.0	647.

```
df.shape
(1041174, 18)

# Changing the data type of fl_date column
df['fl_date'] = pd.to_datetime(df['fl_date'])
df['fl_date'].dtype
dtype('<M8[ns]>')

# Convert categorical columns
cat_cols = ['origin', 'origin_city_name', 'origin_state_nm']
for col in cat_cols:
    df[col] = df[col].astype('category')
```

IMPUTATION

Assigning '0' to the columns taxi_out, taxi_in, Wheels_off, Wheels_on and dep_time when the flight is cancelled i.e '1'

```
cols = ['taxi_out', 'taxi_in', 'wheels_off',
        'wheels_on', 'dep_time', 'air_time']

for col in cols:
    df.loc[df['cancelled'] == 1, col] = 0
```

Removing (',' and city name) from origin_city_name column

```
df['origin_city_name'] = df['origin_city_name'].str.split(',').str[0]  
df
```

...	year	month	day_of_month	day_of_week	fl_date	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	wheels_off	wheels_on
0	2024	1	1	1	2024-01-01	JFK	New York	New York	1247.0	31	1318.0	1442.0
1	2024	1	1	1	2024-01-01	MSP	Minneapolis	Minnesota	1001.0	20	1021.0	1249.0
2	2024	1	1	1	2024-01-01	JFK	New York	New York	1411.0	21	1432.0	1533.0

```
df['dep_time'] = pd.to_datetime(  
    df['dep_time'].astype('Int64').astype(str).str.zfill(4),  
    format='%H%M',  
    errors='coerce'  
)
```

```
df['dep_time'] = df['dep_time'].dt.strftime('%I:%M %p')
```

Note: For the above value after changing the data type of dep_time to hour minutes format, I get 82 null values because it is given as 24.00, which is an invalid time..

```
# Creating a new column to analyse air_time  
  
df['calculated_air_time'] = (((df['wheels_on'] // 100) * 60 + (df['wheels_on'] % 100))  
    - ((df['wheels_off'] // 100) * 60 + (df['wheels_off'] % 100)))
```

Dropping some columns with more inconsistent values that may affect the accuracy.

```
df = df.drop(columns=['wheels_on', 'wheels_off', 'air_time', 'calculated_air_time'],  
    errors='ignore')
```

df	year	month	day_of_month	day_of_week	fl_date	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	taxi_in	cancelled	d
0	2024	1	1	1	2024-01-01	JFK	New York	New York	12:47 PM	31	7	0	
1	2024	1	1	1	2024-01-01	MSP	Minneapolis	Minnesota	10:01 AM	20	6	0	
2	2024	1	1	1	2024-01-01	JFK	New York	New York	02:11 PM	21	8	0	

Inserting a new day column from fl_date column with positioning next to fl_date column

```
df.insert(  
    df.columns.get_loc('fl_date') + 1,  
    'day_name',  
    df['fl_date'].dt.day_name()  
)
```

df	year	month	day_of_month	day_of_week	fl_date	day_name	origin	origin_city_name	origin_state_nm	dep_time	taxi_out	taxi_in	ca
0	2024	1	1	1	2024-01-01	Monday	JFK	New York	New York	12:47 PM	31	7	
1	2024	1	1	1	2024-01-01	Monday	MSP	Minneapolis	Minnesota	10:01 AM	20	6	
2	2024	1	1	1	2024-01-01	Monday	JFK	New York	New York	02:11 PM	21	8	

Inserting a new month_name column and stored it in the 6th index

```
df['month_name'] = df['fl_date'].dt.month_name()
c1m = df.pop('month_name')
df.insert(6, 'month_name', c1m)
```

	year	month	day_of_month	day_of_week	fl_date	day_name	month_name	origin	origin_city_name	origin_state_nm	dep_time	taxi_out
0	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	12:47 PM	31
1	2024	1	1	1	2024-01-01	Monday	January	MSP	Minneapolis	Minnesota	10:01 AM	20
2	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	02:11 PM	21

```
# Renaming the column names 'origin_city_name' and 'origin_state_nm'
df.rename(columns = {'origin_city_name': 'origin_city',
                    'origin_state_nm' : 'origin_state'}, inplace = True)

df
```

	year	month	day_of_month	day_of_week	fl_date	day_name	month_name	origin	origin_city	origin_state	dep_time	taxi_out	taxi_in
0	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	12:47 PM	31	7
1	2024	1	1	1	2024-01-01	Monday	January	MSP	Minneapolis	Minnesota	10:01 AM	20	6
2	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	02:11 PM	21	8

```
q1 = df['ground_time'].quantile(0.33)
q2 = df['ground_time'].quantile(0.66)

df['congestion_level'] = pd.cut(df['ground_time'],
                               bins=[-1, q1, q2, df['ground_time'].max()],
                               labels=['Low', 'Medium', 'High'])

df
```

	year	month	day_of_month	day_of_week	fl_date	day_name	month_name	origin	origin_city	origin_state	...	taxi_out	taxi_in	can
0	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	...	31	7	
1	2024	1	1	1	2024-01-01	Monday	January	MSP	Minneapolis	Minnesota	...	20	6	
2	2024	1	1	1	2024-01-01	Monday	January	JFK	New York	New York	...	21	8	

```
def time_slot(hour):
    if 5 <= hour < 12:
        return 'Morning'
    elif 12 <= hour < 17:
        return 'Afternoon'
    elif 17 <= hour < 21:
        return 'Evening'
    else:
        return 'Night'

df['time_slot'] = pd.to_datetime(df['dep_time']).dt.hour.apply(time_slot)
```



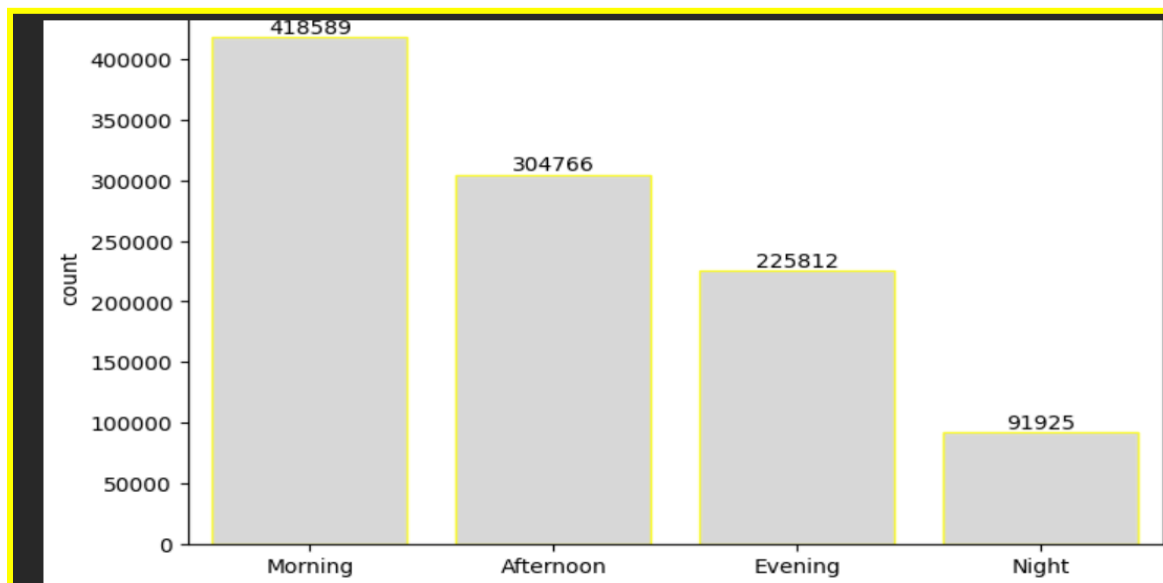
```
# dropping null values in dep_time.
df = df.dropna(subset=['dep_time'])
```

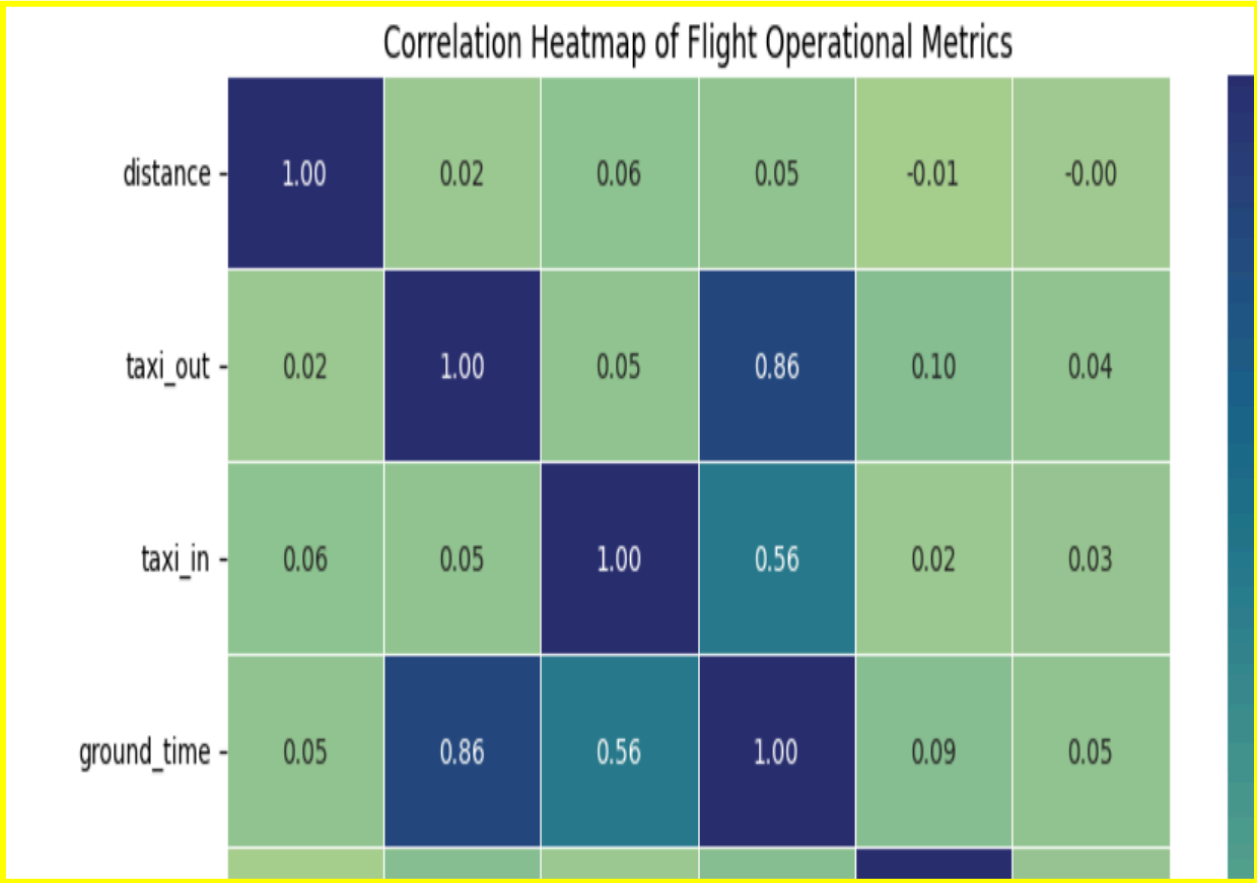
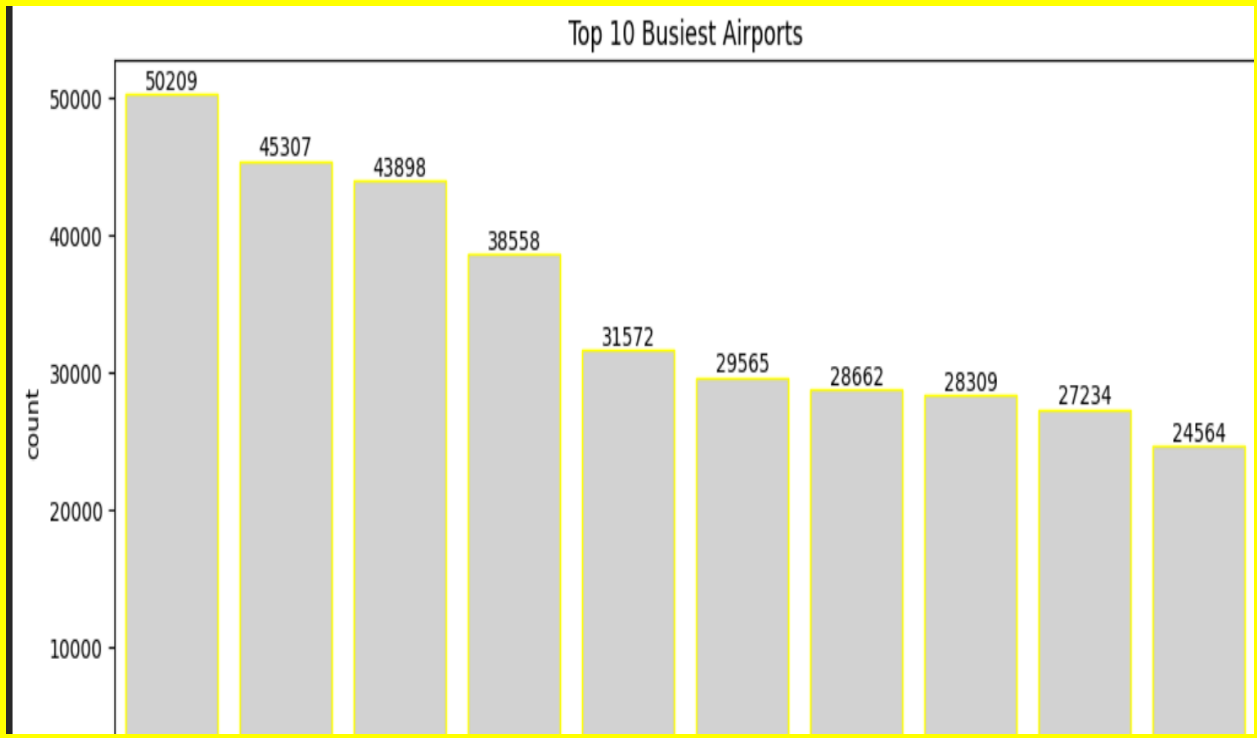
```
df.isnull().sum()
```

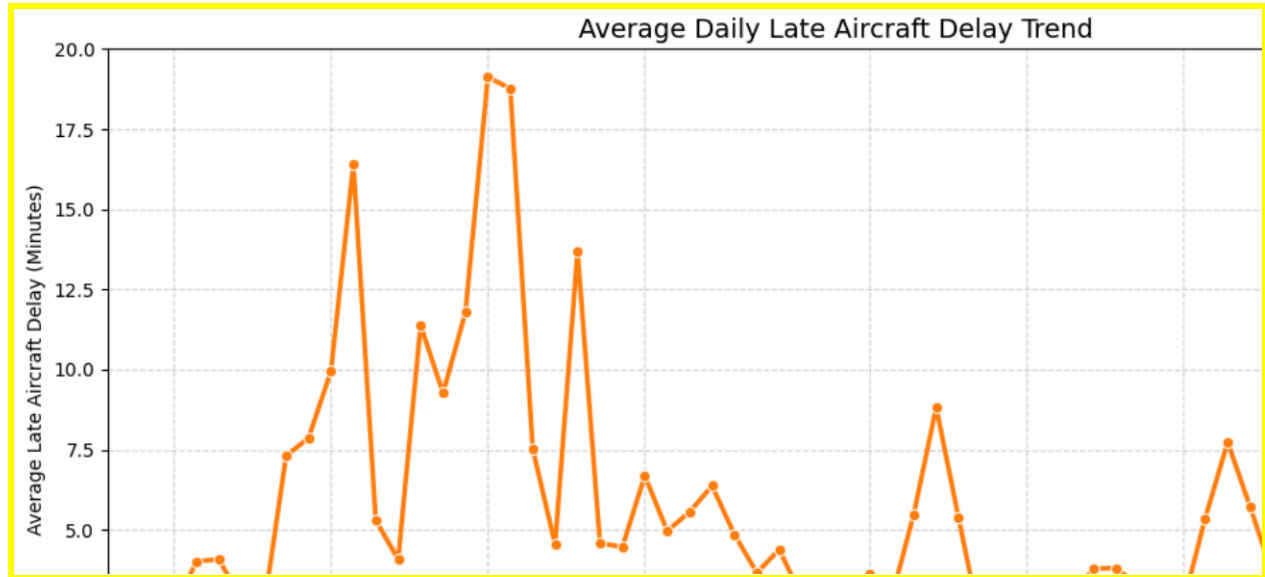
	0
year	0
month	0
day_of_month	0
day_of_week	0
fl_date	0
day_name	0
month_name	0
origin	0
origin_city	0

origin_state	0
dep_time	0
taxi_out	0
taxi_in	0
cancelled	0
distance	0
weather_delay	0
late_aircraft_delay	0
flight_status	0
weather_delay_log	0
ground_time	0
congestion_level	0
time_slot	0

EDA & Visualizations

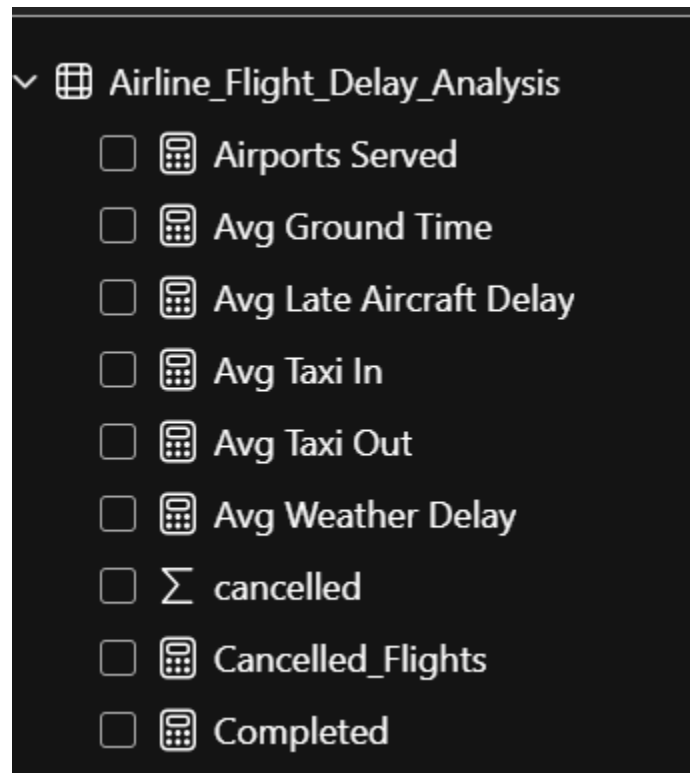






DAX (Power BI)

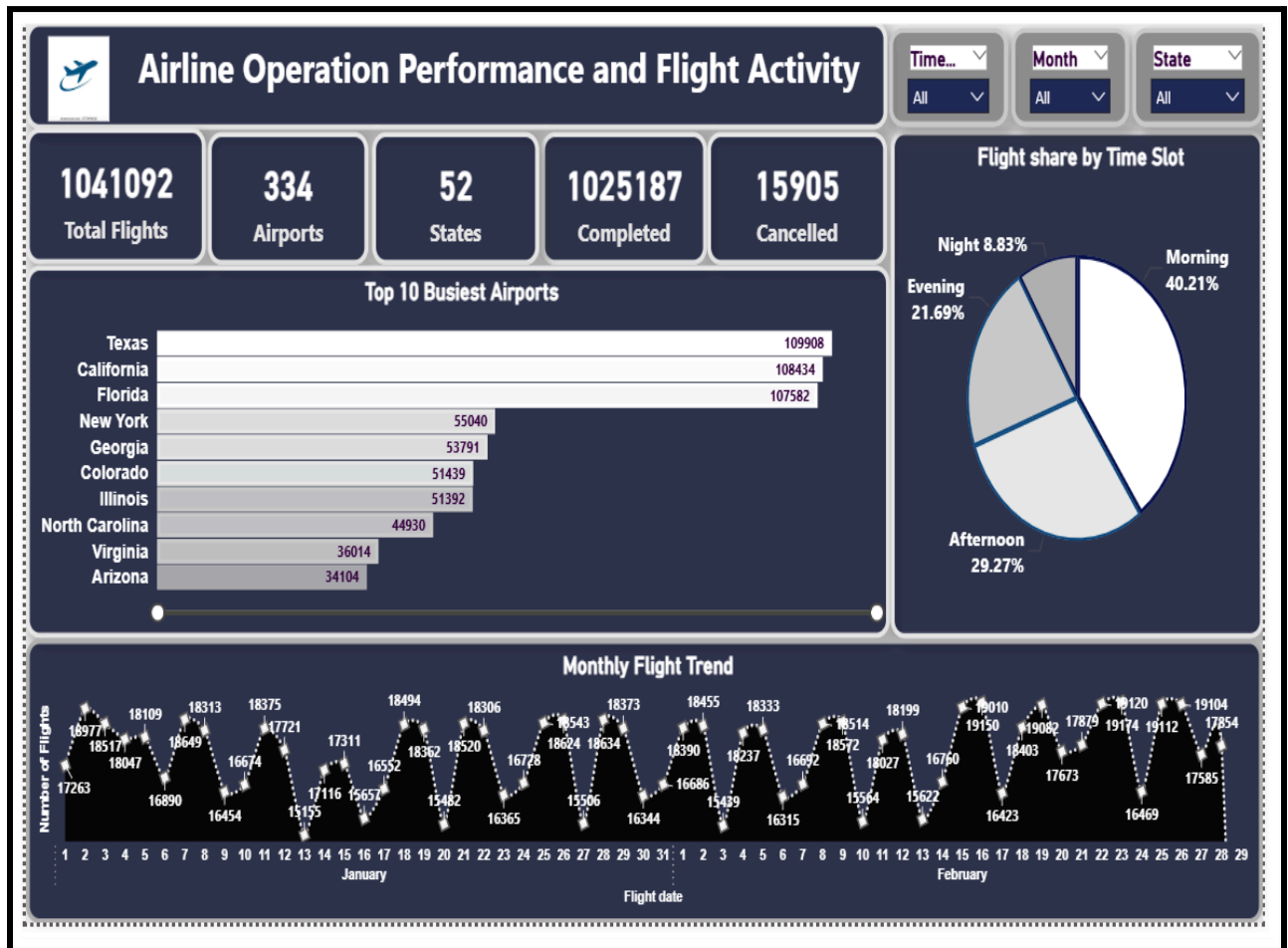
Created DAX measure for the following:



8. Analysis and Visualizations (Power BI) Dashboard Features:

Multiple Visualizations based on problem statements : KPI's, Gauge, Bar charts, Line charts, Pie charts, Area plot, Column chart, Maps and Matrix to communicate insights.

Insight- Dashboard 1: AirLine Operational Performance and Flight Activity



1. KPI Cards

Total Flights (1,041,092)

- More than 1.04 million flights were operated during the analysis period, indicating a high level of airline activity.
- The large flight volume provides a comprehensive view of operational performance across multiple airports and states.

Airports Covered (334)

- Flight operations were distributed across 334 airports, demonstrating extensive network coverage.
- The wide airport reach reflects strong connectivity and geographical diversity.

States Covered (52)

- Airlines operated across 52 states and territories, ensuring broad regional accessibility.
- The network coverage highlights nationwide operational presence.

Completed Flights (1,025,187)

- The majority of flights were successfully completed, indicating high operational reliability.
- Completed flights accounted for over 98% of total operations, showing efficient airline performance.

Cancelled Flights (15,905)

- Only a small percentage of flights were cancelled compared to total operation
- The low cancellation count suggests effective scheduling and operational management.

2. Top 10 Busiest Airports (Horizontal Bar Chart)

- Texas, California, and Florida recorded the highest flight volumes, making them the busiest operational regions.
- Flight activity was highly concentrated among a few major airports, reflecting significant passenger and traffic demand.

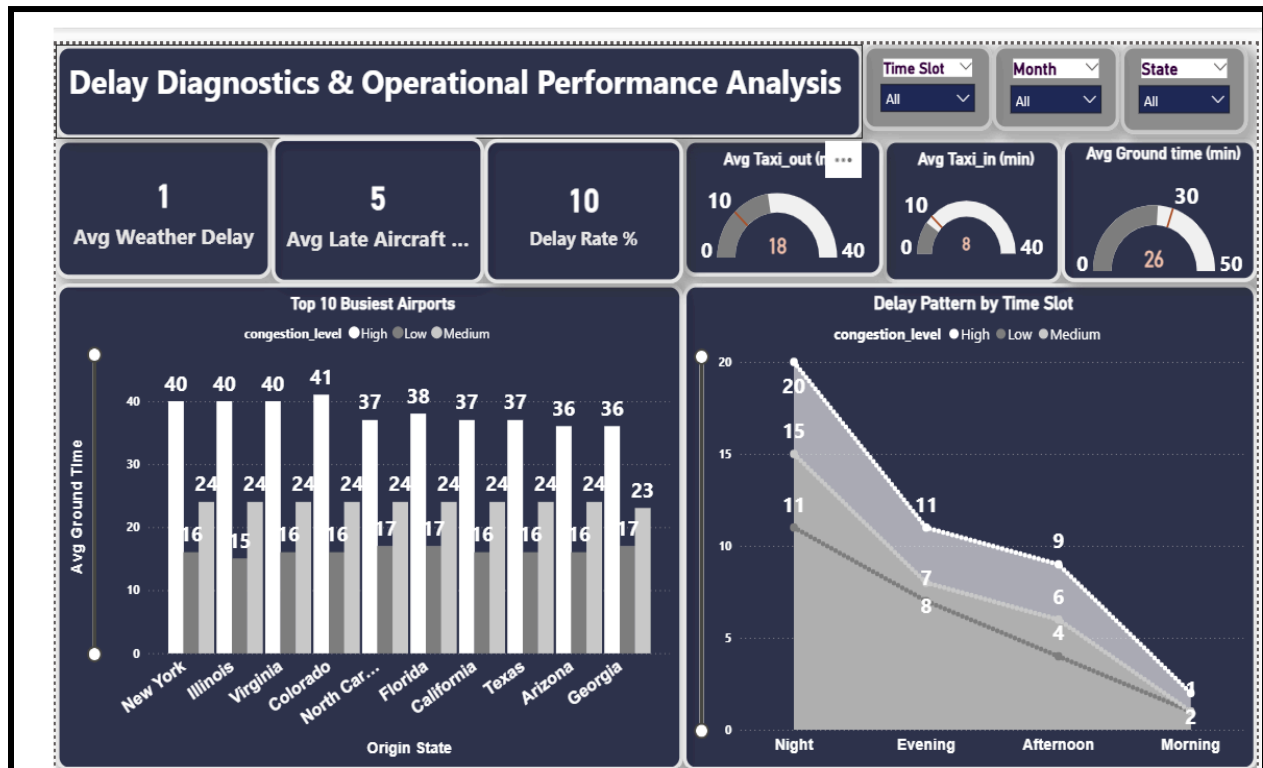
3. Flight Share by Time Slot (Pie Chart)

- Morning flights accounted for the largest share (40.21%) of total operations, indicating peak activity during early hours.
- Night flights represented the smallest proportion (8.83%), suggesting reduced operational demand during late hours.

4. Monthly Flight Trend (Line Chart)

- Daily flight volumes remained relatively stable throughout January and February, showing consistent airline operations.
- Several peaks and dips were observed, indicating variations in travel demand across different days.

Dashboard Insight 2: Delay Diagnostics & Operational Performance Analysis



1. Average Weather Delay (KPI Card)

- Weather-related disruptions had a minimal impact on overall airline operations, with an average delay of only **1 minute**.
- Efficient scheduling and favorable weather conditions helped maintain operational stability.

2. Average Late Aircraft Delay (KPI Card)

- Late-arriving aircraft contributed more significantly to delays than weather-related factors.
- Improving aircraft turnaround efficiency can help reduce cascading delays.

3. Delay Rate % (KPI Card)

- Only around **10%** of flights experienced delays, indicating strong operational reliability.
- Most flights operated according to schedule, reflecting effective airline management.

4. Average Taxi-Out Time Gauge

- The average taxi-out time was approximately **18 minutes**, indicating moderate departure-side congestion.
- Efficient runway and ground movement management are essential to reduce departure delays.

5. Average Taxi-In Time Gauge

- Aircraft spent an average of **8 minutes** taxiing after landing, showing relatively efficient arrival operations.
- Lower taxi-in times contribute to faster gate turnaround and improved operational performance.

6. Average Ground Time Gauge

- Average ground time remained around **26 minutes**, reflecting stable turnaround performance.
- Reducing ground time can enhance aircraft utilization and improve scheduling efficiency.

7. Average Ground Time by State and Congestion Level (Clustered Column Chart)

- States with **High congestion levels** consistently recorded longer average ground times compared to Medium and Low congestion categories.
- New York, Colorado, Florida, and Texas experienced relatively higher ground times, indicating increased airport traffic and operational complexity.

8. Delay Pattern by Time Slot (Line Chart)

- Flights operating during the **Night** time slot experienced the highest delay levels across all congestion categories.
- **Morning flights** recorded the lowest delay values, indicating better operational efficiency during early hours.

Overall Dashboard Findings

- Late aircraft delays had a greater impact on operations than weather-related delays.
- Higher congestion levels were associated with increased ground times and delay occurrences.

- Morning operations demonstrated better schedule adherence, while night operations experienced the highest delays.
- Optimizing aircraft turnaround processes and reducing airport congestion can significantly improve airline operational performance.

Detailed Operational Performance Table

State	Month_name	Average of taxi_in	Average of taxi_out	Avg Late Aircraft Delay	Day Name	Total_Flights
U.S. Pacific Trust Territories and Possessions	February	4.00	13.67	0.00	Saturday	12
U.S. Pacific Trust Territories and Possessions	February	4.08	12.42	0.00	Wednesday	12
U.S. Pacific Trust Territories and Possessions	February	4.25	13.58	0.00	Sunday	12
U.S. Pacific Trust Territories and Possessions	February	4.33	13.33	0.00	Friday	12
U.S. Pacific Trust Territories and Possessions	January	4.58	13.75	0.00	Saturday	12
U.S. Pacific Trust Territories and Possessions	January	4.67	13.42	0.00	Sunday	12
U.S. Pacific Trust Territories and Possessions	February	4.75	14.67	0.00	Tuesday	12
U.S. Pacific Trust Territories and Possessions	January	5.08	16.50	0.00	Friday	12
U.S. Pacific Trust Territories and Possessions	January	5.27	13.20	0.00	Tuesday	15
U.S. Pacific Trust Territories and Possessions	February	5.75	15.31	3.00	Monday	16
Alaska	January	5.79	14.26	10.00	Saturday	306
Alaska	February	5.93	15.84	2.00	Saturday	314
Alaska	January	6.10	17.24	5.00	Tuesday	384
U.S. Pacific Trust Territories and Possessions	January	6.10	12.50	20.00	Monday	20
Total		7.96	17.97	5.00		1041092

Flight Status
All

Day
All

Origin State
All

Flight Date
All

congestion_level
☐ High
☐ Low

1. State-wise Operational Metrics

- Significant variations exist in average taxi-in and taxi-out times across states, indicating differences in airport congestion and operational efficiency.
- States with higher flight volumes generally exhibit longer ground movement times due to increased traffic density.

2. Monthly Performance Comparison

- Operational metrics remain relatively stable between January and February, suggesting consistent airline performance across both months.
- Minor fluctuations in taxi-in and taxi-out times indicate seasonal and traffic-related variations in airport operations.

3. Day-wise Flight Activity

- Flight volumes and operational performance vary across different days of the week, highlighting the impact of travel demand patterns.
- Weekends and high-demand days tend to experience greater traffic and slightly increased ground operation times.

4. Average Taxi-In Time

- The overall average taxi-in time of approximately 8 minutes reflects efficient arrival-side ground operations.
- Lower taxi-in times contribute to improved turnaround efficiency and better gate utilization.

5. Average Taxi-Out Time

- Average taxi-out time of around 18 minutes indicates moderate departure-side congestion at airports.
- Managing runway traffic and ground movement can help reduce departure delays and improve operational efficiency.

6. Average Late Aircraft Delay

- Late aircraft delays remain relatively low on average, demonstrating effective aircraft scheduling and turnaround management.
- However, certain states and days experience higher delay values, suggesting localized operational challenges.

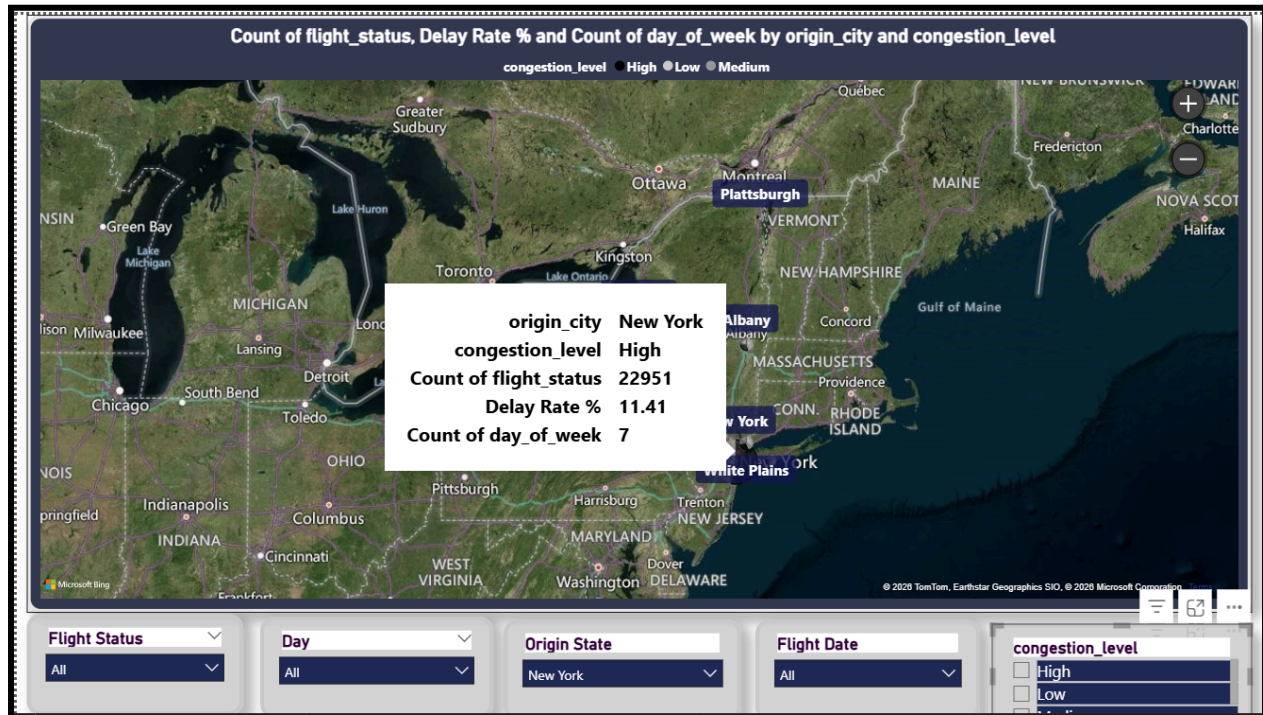
7. Total Flight Volume

- More than 1.04 million flights were analyzed, providing a comprehensive view of airline operations.
- High flight volumes across multiple states demonstrate extensive network coverage and robust airline activity.

Overall Findings

- Airport congestion significantly influences taxi-in and taxi-out times.
- Operational performance remained relatively consistent throughout the analysis period.
- Flight activity patterns differ across states, months, and days of the week.
- Efficient ground operations and aircraft scheduling play a crucial role in minimizing delays and improving overall airline performance.

Geographical Flight Activity & Delay Distribution Map



Insights

1. Geographic Coverage of Flight Operations

- Flight operations are widely distributed across the United States, demonstrating extensive network connectivity and nationwide airline coverage.
- Major metropolitan regions and transportation hubs exhibit higher concentrations of flight activity compared to less populated areas.

2. City-wise Flight Volume

- Cities such as Dallas, Chicago, Atlanta, New York, Los Angeles, and Minneapolis represent major centers of airline operations.
- Higher flight volumes are concentrated around economically significant and densely populated regions.

3. Delay Rate Distribution

- Delay rates vary across different cities, indicating that operational efficiency is influenced by local airport conditions and traffic density.
- Airports located in high-traffic regions are more susceptible to operational disruptions and congestion-related delays.

4. Congestion Level Analysis

- The presence of high, medium, and low congestion categories highlights variations in airport traffic intensity across the network.
- High congestion regions tend to experience increased ground operations and greater delay frequencies.

5. Spatial Pattern of Airline Operations

- Airline activity is concentrated in strategic transportation corridors, reflecting the importance of major airports in supporting national connectivity.
- Flight density gradually decreases in remote and less populated regions, indicating lower operational demand.

6. Operational Performance Across Regions

- Regional differences in congestion levels and delay rates suggest opportunities for improving airport resource allocation and traffic management.
- Understanding spatial patterns enables airlines to optimize scheduling and enhance overall operational efficiency.

Overall Findings

- Airline operations exhibit broad geographical coverage with activity concentrated around major urban and economic centers.
- Congestion and delay patterns vary significantly by location, emphasizing the impact of airport traffic conditions on operational performance.
- Major hubs play a critical role in maintaining nationwide connectivity but are also more vulnerable to congestion-related disruptions.
- Geospatial analysis provides valuable insights for improving route planning, airport management, and operational efficiency.

Future Enhancements

- **Real-time flight and weather data will be integrated** to enable live operational monitoring and analysis.
- **Machine learning models will be developed** to predict flight delays and identify potential disruptions in advance.
- **Additional features such as airline carriers, destination airports, and seasonal trends will be incorporated** to provide deeper operational insights.
- **Advanced interactive dashboards and automated reporting capabilities will be implemented** to enhance decision-making and user experience.

Conclusion:

This project analyzed over 1 million flight records to evaluate airline operational performance and identify key factors influencing delays and congestion. Through data cleaning, feature engineering, and exploratory analysis, valuable insights were obtained regarding flight activity, airport efficiency, ground operations, and delay patterns.

Interactive Power BI dashboards enabled effective visualization of operational metrics, revealing that late aircraft delays and airport congestion have a greater impact on performance than weather-related disruptions. The findings provide actionable insights that can support improved scheduling, resource allocation, and overall airline operational efficiency.